# HELM & HELM CHART

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## What is Helm

1. Helm it's a tool that automates the creation, packaging, configuration, deployment of k8s application by combining your configuration files into a single reusable package.
2. helm chart helps you to define, install, even upgrade the most complex Kubernetes applications.

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### INSTALLATION OF HELM

$ curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-4
$ chmod 700 get_helm.sh
$ ./get_helm.sh
$ helm version
{ v4.1.0 }

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### CREATING OUR OWN HELM CHART ==>> (APACHE)

1. helm create apache-helm
2. cd apache-helm
3. ls
   chart.yaml charts/ templates/ values.yaml

*All manifests' files are stored in the templates/*

1. deployment.yaml
2. service.yaml
3. ingress.yaml
4. hpa.yaml

*All values are defined in* values.yaml

```
replicaCount: 2
image:
  repository: httpd
  tag: "2.4"
service:
  type: NodePort
  port: 80
  targetPort: 80
  nodePort: 30080
```

*Change in the templates/service.yaml file*

add nodeport section below name section

```
  {{- if eq .Values.service.type "NodePort" }}
  nodePort: {{ .Values.service.nodePort }}
  {{- end }}
```

*All info about the packages are stored in chart.yaml*

1. To detect YAML syntax errors and template issues before deployment.
   helm lint apache-helm

2. Preview the Generated Manifests
   helm template dev-apache apache-helm

3. Generates a compressed .tgz file that can be shared or uploaded to a Helm repository
   helm package apache-helm

4. To deploy the application into the Kubernetes cluster.
   helm install dev-apache apache-helm

5. kubectl get pods
6. kubectl get svc
7. kubectl get deployments

8. To deploy the application into the Kubernetes cluster.
   helm upgrade dev-apache apache-helm

9. To see all Helm releases installed in the current namespace.
   helm list

10. To check the history of revisions
    helm history dev-apache

11. To inspect the current state of a Helm release.
    helm status dev-apache

12. To check which values were used during installation or upgrade.
    helm get values dev-apache

**TO ROLL-BACK**

1. Rollback to revision 1
   helm rollback dev-apache 1

2. To check all the history of dev-apache revision
   helm history dev-apache

3. To remove the application from the Kubernetes cluster.
   helm uninstall dev-apache

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**ASSIGN A NAMESPACE TO A CHART**

1) helm install dev-apache apache-helm -n dev --create-namespace
2) kubectl get pods -n dev

3) helm install prod-apache apache-helm -n prod --create-namespace
4) kubectl get pods -n prod

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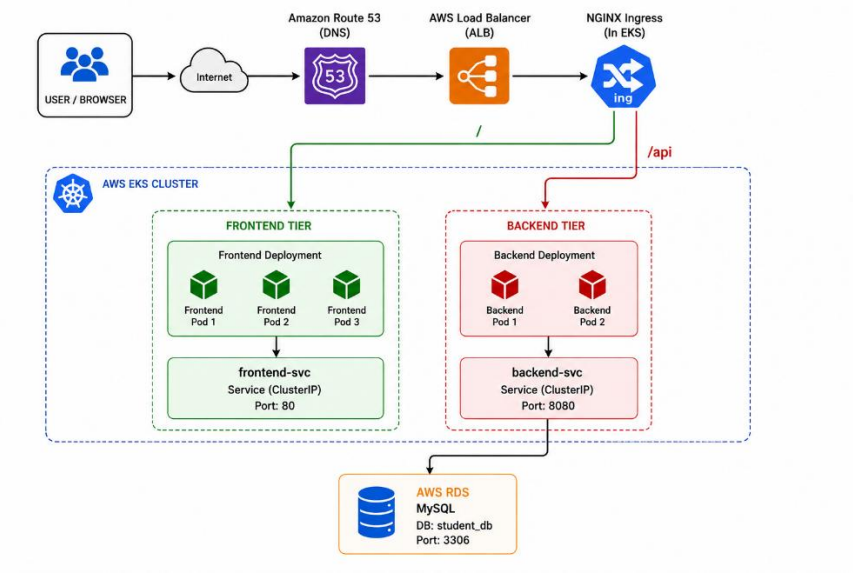
## DEPLOY STUDENTAPP APPLICATION

1. helm create studentapp
2. cd studentapp
3. vim templates/service.yaml
   add one line after targetport {{ .Values.service.targetPort }}
4. vim values.yaml
   replicaCount: 2
   repository: mayurpandit25/student-app
   tag: "latest"
   port: 8080
   targetPort: 8080
5. cd ..
6. helm package studentapp
7. helm install dev-studentapp studentapp -n dev-student --create-namespace
8. kubectl get pods -n dev-student
9. kubectl get svc -n dev-student
10. kubectl port-forward svc/dev-studentapp 8080:8080 --address=0.0.0.0 -n dev-student

<http://<public_ip:8080>

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# DEPLOY THREE-TIER APPLICATION USING HELM (EASY-CRUD)




1.create a RDS (MySQL) database

## ON BACKEND

1. access the server
    1. apt update
    2. apt install mysql-client
    3. mysql -u admin -h <endpoint> -p
    4. create database student_db;

2. git clone https://github.com/mayurpandit25/EasyCRUD.git
3. cd EasyCRUD/backend/src/main/resources
4. vim application.properties
    1. Endpoint:
    2. User:
    3. Password:
5. docker build -t mayurpandit25/cbz-backend:latest .
6. docker push mayurpandit25/cbz-backend:latest

## ON FRONTEND

1. cd frontend
2. vim .env
   VITE_API_URL=/api
3. docker build -t mayurpandit25/cbz-frontend:latest
4. docker push mayurpadit25/cbz-frontend:latest

## USING HELM

1. helm create easycrud
2. cd easycrud

3. cd templates
4. rm -rf *
5. vim Chart.yaml
   apiVersion: v2
   name: easycrud
   description: Three Tier Application Helm Chart
   type: application
   version: 1.0.0
   appVersion: "1.0"

6. vim values.yaml
   backend:
    replicaCount: 2

    image:
     repository: mayurpandit25/cloudblitz-backend
     tag: v1
     pullPolicy: Always

    service:
     type: ClusterIP
     port: 8080
     targetPort: 8080

   frontend:
    replicaCount: 1

    image:
     repository: mayurpandit25/cloudblitz-frontend
     tag: v1
     pullPolicy: Always

    service:
     type: ClusterIP
     port: 80
     targetPort: 80

   ingress:
    enabled: true
    className: nginx

    host: app.mayurcbz.space

    frontendPath: /
    backendPath: /api

7. vim backend-deployment.yaml
   apiVersion: apps/v1
   kind: Deployment
   metadata:

```
  name: backend
spec:
  replicas: {{ .Values.backend.replicaCount }}
  selector:
    matchLabels:
      app: backend
  template:
    metadata:
      labels:
        app: backend
    spec:
      containers:
        - name: backend
          image: "{{ .Values.backend.image.repository }}:{{ .Values.backend.image.tag }}"
          imagePullPolicy: {{ .Values.backend.image.pullPolicy }}
          ports:
            - containerPort: {{ .Values.backend.service.targetPort }}
```

8. vim backend-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: backend-svc
spec:
  type: {{ .Values.backend.service.type }}
  selector:
    app: backend
  ports:
    - port: {{ .Values.backend.service.port }}
      targetPort: {{ .Values.backend.service.targetPort }}
```

9. vim frontend-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend
spec:
  replicas: {{ .Values.frontend.replicaCount }}
  selector:
    matchLabels:
      app: frontend
  template:
    metadata:
      labels:
        app: frontend
    spec:
      containers:
```

```
        - name: frontend
          image: "{{ .Values.frontend.image.repository }}:{{ .Values.frontend.image.tag }}"
          imagePullPolicy: {{ .Values.frontend.image.pullPolicy }}
          ports:
            - containerPort: {{ .Values.frontend.service.targetPort }}
```

10. vim frontend-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: frontend-svc
spec:
  type: {{ .Values.frontend.service.type }}
  selector:
    app: frontend
  ports:
    - port: {{ .Values.frontend.service.port }}
      targetPort: {{ .Values.frontend.service.targetPort }}
```

11. vim ingress.yaml

```
{{- if .Values.ingress.enabled }}
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: three-tier-ingress
spec:
  ingressClassName: {{ .Values.ingress.className }}
  rules:
    - host: {{ .Values.ingress.host }}

      http:
        paths:
          - path: {{ .Values.ingress.backendPath }}
            pathType: Prefix
            backend:
              service:
                name: backend-svc
                port:
                  number: {{ .Values.backend.service.port }}
          - path: {{ .Values.ingress.frontendPath }}
            pathType: Prefix
            backend:
              service:
                name: frontend-svc
                port:
                  number: {{ .Values.frontend.service.port }}
{{- end }}
```

**INSTALL INGRESS**

1. kubectl apply -f https://raw.githubusercontent.com/kubernetes/ingress-nginx/main/deploy/static/provider/aws/deploy.yaml
2. kubectl get pods -n ingress-nginx
3. kubectl get svc -n ingress-nginx

**START DEPLOYMENT**

1. helm lint easycrud
2. helm template studentapp easycrud
3. helm package easycrud
4. helm install studentapp easycrud
5. helm list

6. kubectl get pods
7. kubectl get svc
8. kubectl get ingress

**URLS**

1. http://app.mayurcbz.space
2. http://app.mayurcbz.space/api

**TO INCREASE REPLICAS**

1.helm upgrade studentapp easycrud –set backend.replicaCount=5 –set frontend.replicaCount=5

**CHANGE DOCKER IMAGE USING HELM**

1. helm upgrade studentapp easycrud –set backend.image.tag=v2

   *Helm update version from v1 to v2*

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**USE A SEPARATE PRODUCTION VALUES FILE**

**DEV-ENVIRONMENT**

**1.vim values-dev.yaml**

```
backend:
  replicaCount: 1

frontend:
  replicaCount: 1
```

2.helm install studentapp . -f values-dev.yaml -n dev –create-namespace

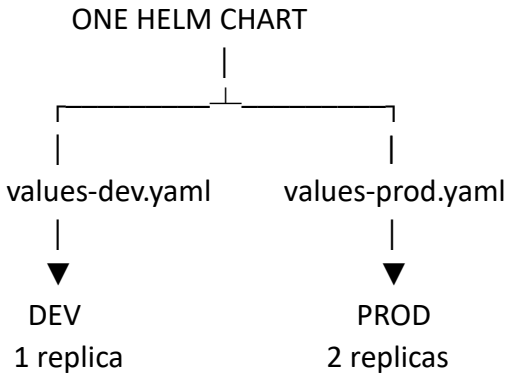
**PROD-ENVIRONMENT**

**1.vim values-prod.yaml**

```
backend:
  replicaCount: 2

frontend:
  replicaCount: 2
```

2.helm install studentapp . -f values-prod.yaml -n prod –create-namespace




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**UPGRADE THE APPLICATION**

2. helm upgrade studentapp easycrud
3. helm list
4. helm history studentapp

**ROLLBACK**

1. helm rollback studentapp 1

**UNINSTALL**

2. helm uninstall studentapp easycrud

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